

Aarav Sharma Student Portfolio

Interactive React + TypeScript Web Application



Project Overview

A modern, fully responsive student portfolio website showcasing academics, projects, skills, certifications, and testimonials. Built with React 19, TypeScript, and Vite, featuring advanced animations powered by GSAP, Three.js, and interactive 3D elements.

Project Goal

Create an engaging, visually stunning personal portfolio that demonstrates technical expertise through a combination of modern web technologies, complex animations, and interactive user experiences.



Technology Stack

Core Framework & Build

React 19.1.0

Modern UI framework with latest features

TypeScript 5.8.0

Type safety and better developer experience

Vite 6.3.0

Lightning-fast build tool with HMR

React DOM 19.1.0

DOM rendering library for React

Styling & Design

Tailwind CSS 4.1.0

Utility-first CSS framework

PostCSS 8.x

CSS processing and optimization

Radix UI

Accessible, unstyled UI components

Shadcn/ui

Copy-paste component library

Animation & 3D Graphics

GSAP 3.13.0

Professional animation library with ScrollTrigger

Three.js 0.175.0

3D graphics engine

@react-three/fiber 9.1.0

React renderer for Three.js

@react-three/drei 10.0.0

R3F helpers and utilities

Physics & Interactions

@react-three/cannon 6.6.0

Physics engine for 3D simulations

Lenis 1.3.0

Smooth scrolling library

Lucide React 0.510.0

Beautiful icon library

@gsap/react 2.1.2

React bindings for GSAP



Dependencies Summary

Runtime Dependencies

Package	Version	Purpose
react	^19.1.0	UI framework
react-dom	^19.1.0	React DOM renderer
three	^0.175.0	3D graphics engine
@react-three/fiber	^9.1.0	React Three.js renderer
@react-three/drei	^10.0.0	R3F helpers and utilities
@react-three/cannon	^6.6.0	Physics engine bindings
gsap	^3.13.0	Animation engine
lenis	^1.3.0	Smooth scrolling
lucide-react	^0.510.0	Icon library

Development Dependencies

Package	Version	Purpose
typescript	^5.8.0	Type safety
vite	^6.3.0	Build tool and dev server
@vitejs/plugin-react	^4.5.0	React Fast Refresh for Vite
tailwindcss	^4.1.0	CSS utility framework
@tailwindcss/vite	^4.1.0	Tailwind Vite integration

Package	Version	Purpose
@types/react	^19.1.0	React type definitions
@types/react-dom	^19.1.0	ReactDOM type definitions
@types/three	^0.175.0	Three.js type definitions



Component Architecture

Layout Components

Header

Fixed navigation with frosted glass effect and scroll-aware transparency

Footer

Centered footer section for contact and navigation

SmoothScrollProvider

App-wide Lenis integration for smooth scrolling

Page Sections (10+ Interactive Sections)

HeroSection

Split-screen layout, animated name, stats counter, particle canvas

AboutSection

Two-column layout with photo and interest pills

AcademicSection

Scroll flip reveal title, grade table display

ClubsSection

Scroll flip reveal, 2-column club cards

CompetitionsSection

Scroll flip reveal title, vertical timeline

ProjectsSection

Dark inset cards, scroll flip reveal, 3D flip card grid

SkillsSection

Scroll flip reveal, 3-column skill cards with progress bars

CertificatesSection

Scroll flip reveal, smooth scroll gallery

BlogSection

Scroll flip reveal, 3-column blog cards

TestimonialsSection

Scroll flip reveal, 2-column testimonial cards

ContactSection

Two-column layout with contact info and form

Reusable Components (15+ Custom)

Component	Used In	Technology
ParticleCanvas	HeroSection	React Three Fiber + Physics
ScrollFlipReveal	Academic, Clubs, Competitions, Projects, Skills, Certificates, Blog, Testimonials	GSAP + SplitText
FlipCard	ProjectsSection	CSS 3D Transforms
SmoothScrollGallery	CertificatesSection	GSAP + Draggable
SectionOverline	All Sections	Custom React Component
SectionHeading	All Sections	Custom React Component
StatCounter	HeroSection	GSAP Animation
PillBadge	About, Competitions, Skills	Custom React Component



Animation Implementation

Key Animations

Animation	Library	Complexity	Technique
Hero Name Entrance	GSAP	Medium	translateY(100%)→0 with overflow:hidden clip, staggered lines
Stats Counter	GSAP	Low	Snap effect for counting animation
Particle Simulation	R3F + Physics	High	55 instanced spheres with Cannon.js physics
Scroll Flip Reveal	GSAP + SplitText	High	Character-level 3D rotation from rotateY(-80°) to 0
CSS 3D Flip Cards	Pure CSS	Medium	perspective + preserve-3d, rotateY(180deg) on hover
Smooth Scroll Gallery	GSAP + Draggable	High	Drag-based horizontal scroll with quickTo interpolation
Section Entrance	GSAP + ScrollTrigger	Low	translateY(30px)→0, opacity 0→1, staggered
Header Scroll Transition	CSS	Low	backdrop-filter and border toggle at 100vh



Note on GSAP Premium Plugins: SplitText, Draggable, and InertiaPlugin require GSAP Club membership. Fallbacks using open-source alternatives (custom character splitting, manual velocity-based drag) are recommended for production without membership.



Project Metrics

10+

Interactive Sections

15+

Custom
Components

8+

Animation Types

25+

Radix UI
Components



Project Structure

```
portfolio/
├─ app/
│  ├─ src/
│  │  ├─ components/
│  │  │  ├─ ScrollFlipReveal.tsx
│  │  │  └─ ui/ (25+ Radix UI components)
│  │  ├─ hooks/
│  │  │  ├─ use-mobile.ts
│  │  │  └─ useSmoothScroll.tsx
│  │  ├─ lib/
│  │  │  └─ utils.ts
│  │  ├─ pages/
│  │  │  └─ Home.tsx
│  │  ├─ sections/ (11 section components)
│  │  ├─ App.tsx
│  │  ├─ App.css
│  │  ├─ main.tsx
│  │  └─ index.css
│  ├─ public/
│  ├─ package.json
│  ├─ vite.config.ts
│  ├─ tsconfig.json
│  ├─ tailwind.config.js
│  └─ eslint.config.js
└─ tech-spec.md
```

Key Features & Highlights

Interactive Animations

- **Scroll-Triggered Effects:** Animations linked to scroll position using GSAP ScrollTrigger
- **3D Particle System:** Physics-based particle simulation in hero section
- **Character-Level Text Animation:** Individual character rotation with flip effect
- **Smooth Scrolling:** Lenis integration for enhanced scroll experience
- **Drag Interactions:** Gallery support for drag-based navigation

Modern Web Technologies

- **React 19:** Latest React features and improvements
- **TypeScript:** Full type safety throughout the application
- **Vite:** Ultra-fast build tool with instant HMR
- **Tailwind CSS:** Utility-first CSS with custom configuration
- **Radix UI:** Accessible, unstyled component primitives

Responsive Design

- Mobile-first approach using Tailwind CSS
- Flexible grid layouts for different screen sizes
- Touch-friendly interactions for mobile devices
- Performance optimized for all devices

Technical Decisions & Rationale

Why React 19 + TypeScript?

React 19 provides the latest features and performance improvements, while TypeScript ensures type safety across a complex component hierarchy. This combination improves developer productivity and reduces runtime errors.

Why Vite?

Vite offers exceptional build performance with instant HMR, essential for rapid development iteration. The dev server startup is near-instantaneous, improving developer experience significantly.

Why GSAP for Animations?

GSAP is the industry-standard animation library with superior performance, ease of use, and ScrollTrigger plugin for scroll-linked animations. The learning curve is minimal for developers familiar with animation libraries.

Why Three.js + React Three Fiber?

Three.js provides robust 3D graphics capabilities, while React Three Fiber seamlessly integrates Three.js with React's component model. This approach provides better code organization and reusability compared to vanilla Three.js.

Why Tailwind CSS?

Tailwind's utility-first approach enables rapid development with consistency. Custom configuration supports the design system, and the small bundle size keeps the application lightweight.



Interview Talking Points

Animation Performance & Optimization

Discuss how GSAP and ScrollTrigger are optimized for performance with GPU-accelerated transforms. Explain the approach to managing complex animations without impacting page responsiveness. Mention the use of will-change CSS properties and transform-based animations.

3D Graphics & Physics Integration

Explain the particle system implementation using Three.js, React Three Fiber, and Cannon.js. Discuss challenges in managing 3D graphics within a React application and how React Three Fiber solves these problems.

Responsive Design at Scale

Address how the portfolio handles animations and complex layouts across different screen sizes. Explain the mobile-first approach and the use of Tailwind's responsive utilities.

TypeScript Benefits

Highlight how TypeScript catches errors during development, improves IDE autocomplete for complex animation APIs, and makes refactoring safer when managing intricate component hierarchies.

Component Reusability

Discuss the creation of reusable animated components (ScrollFlipReveal, StatCounter, ParticleCanvas) and how they reduce code duplication across the 10+ sections.

Accessibility & User Experience

Mention the use of Radix UI for accessible component primitives. Discuss considerations for animations and how they enhance rather than hinder the user experience.

Build Tool Selection

Explain why Vite was chosen over alternatives like Webpack or Create React App. Discuss the developer experience improvements and faster build times.

Smooth Scrolling & Performance

Discuss the Lenis library for smooth scrolling and how it works alongside scroll-triggered animations without impacting performance.



Conclusion

This portfolio project demonstrates modern web development practices with a focus on interactive design, performance optimization, and code quality. It showcases expertise in React, TypeScript, advanced CSS techniques, 3D graphics, and complex animations. The project serves as both a personal portfolio and a technical showcase of skills in building engaging, high-performance web applications.

The combination of cutting-edge technologies (React 19, Vite, Three.js) with industry-standard animation libraries (GSAP) results in a visually stunning and performant application that effectively communicates technical capabilities to potential employers and collaborators.

Project Documentation • Generated on 2026-06-15

Technology Stack: React 19 | TypeScript | Vite | GSAP | Three.js | Tailwind CSS